



AI & prompt governance

TASCO Insurance × VETC Growth Platform — Architecture

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Scope: every automated decision or AI component that affects customers on the platform.

- The voice bot: dialogue policy, intent classifier, and the vendor's ASR/TTS.
- The lead score, next-best action and journey assignment.
- MDM inference of policy expiry and vehicle category.
- Any future LLM or ML model.

Related: [ADR-003](#), [ADR-009](#). Regulatory statements are flagged "**confirm with TASCO legal**".

1. AI inventory and risk tiering

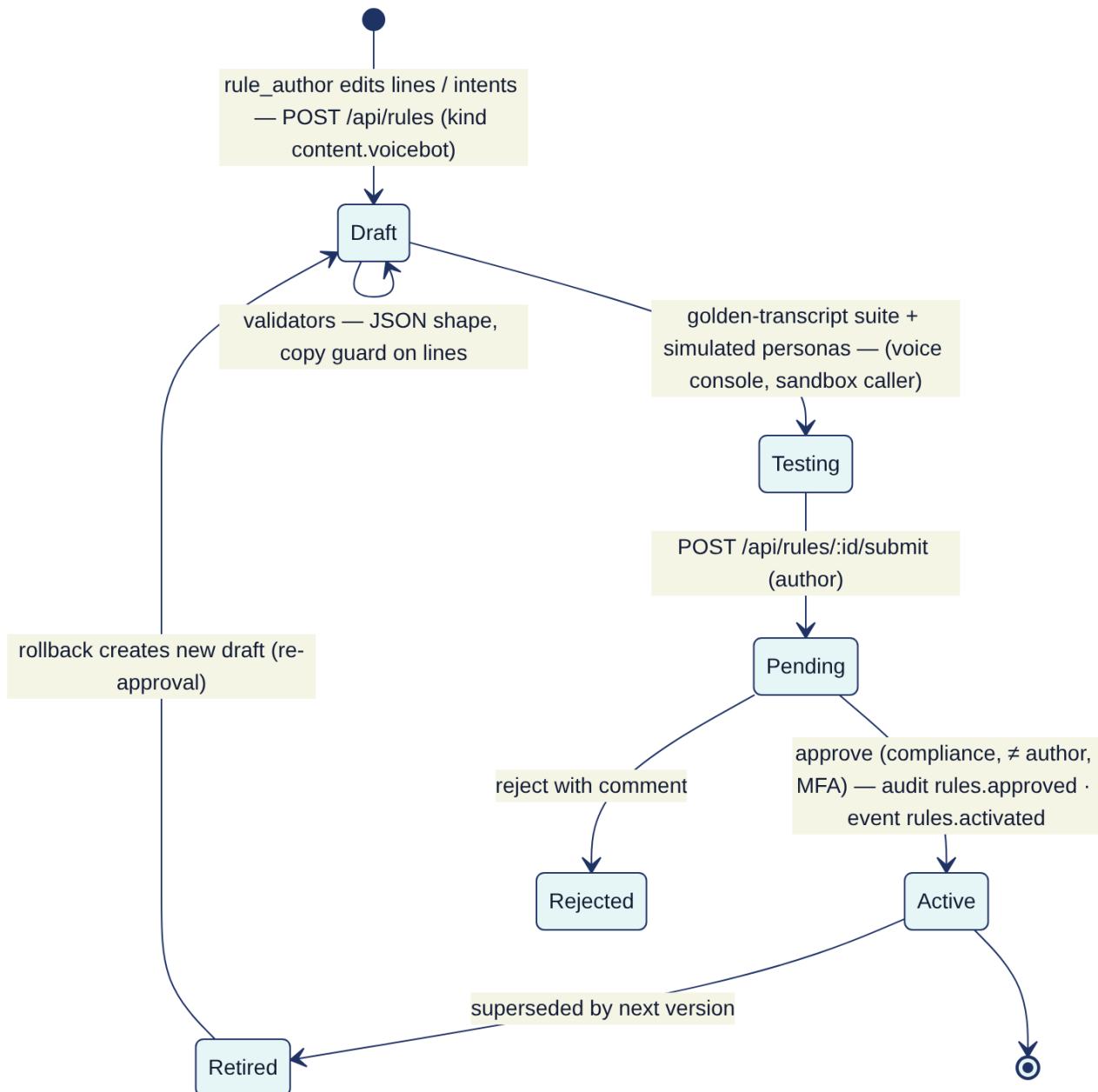
#	Component	Type	Code / config	Customer impact	Risk tier	Owner
1	Voice bot dialogue	Deterministic state machine + governed script	<code>src/domain/voicebot.js</code> , <code>config/rules/content.voicebot.json</code>	Speaks to customers; captures opt-out; hands off to sales	High (direct interaction, vishing context)	CX lead
2	Intent classifier	Keyword NLU (default); LLM optional (future)	<code>keywordClassifier</code> in <code>voicebot.js</code>	Wrong intent leads to a wrong path (a missed opt-out is critical)	High	CX lead + model risk
3	ASR / TTS	Vendor ML (Vietnamese)	Telephony port (Integration §3.5)	Mis-transcription → plate mismatch or wrong intent	Medium	Vendor + CX
4	Lead score + tier	Explainable weighted rules (white-box)	<code>src/domain/leads.js#score</code> , <code>config/rules/scoring.json</code>	Who gets contacted, how often, on which channel	Medium	Growth analytics
5	Next-best action, journey assignment	Decision tables / JSON Logic	<code>nba.json</code> , <code>journeys.json</code> , <code>leads.js</code>	Channel choice, suppression, B2B routing	Medium	Campaign
6	Expiry inference	Evidence-weighted inference	<code>src/domain/enrichment.js#inferExpiry</code> , <code>enrichment.json</code>	Timing of reminders; quote start date	Medium	Data office
7	Vehicle category inference	Decision table	<code>enrichment.categoryTable</code>	Regulated premium amount if not confirmed	High for pricing correctness	Data office + underwriting
8	Benefits relevance	JSON Logic relevance scores	<code>benefits.json</code> , <code>leads.js#benefits</code> For	Which value-adds are shown (only <code>legalStatus=approved</code> items to customers)	Low	Product
9	Future: LLM classifier / agent assist; propensity model	ML/LLM	Not implemented	—	Assessed before introduction	Model risk

2. Voice bot design principles

Each principle is enforced in code or governed content.

#	Principle	How it is enforced
1	Disclose automation and recording at call start	The <code>intro</code> line: "đây là trợ lý tự động của VETC ... Cuộc gọi được ghi âm" ("this is VETC's automated assistant ... this call is recorded"). Payload <code>disclosure: "automated_assistant"</code> . Governance dashboard states the disclosure. Recording-notice wording: confirm with TASCO legal .
2	Never ask for an OTP, card or payment on a call	Stated in <code>intro</code> and the <code>trust</code> line. The dialogue has no state that collects such data. Payment happens only in the VETC app (<code>link</code> , <code>handoff</code> lines).
3	Plate-first verification. The customer says the plate; the bot never reads it out.	State <code>verify_plate: extractPlateFromSpeech</code> must equal the profile key. A mismatch ends the call (<code>plate_mismatch</code>), and after <code>maxPlateAttempts</code> (3) the call ends <code>unverified</code> . Afterwards the bot refers to the vehicle only by <code>plateMasked</code> (<code>30A-***.45</code>).
4	Truthful about price. The premium is regulated and identical everywhere; no discounts.	The <code>price</code> line states the regulated premium incl. VAT and pivots to service value. The copy guard bans <code>giảm giá</code> , <code>chiết khấu</code> , <code>hoàn tiền</code> , <code>discount</code> , <code>cashback</code> , <code>rebate</code> , <code>cheaper</code> , ... (<code>copy_guard.json</code>). Validation fails if a line contains them.
5	Respect opt-out immediately	<code>opt_out</code> is a global intent checked before any state logic. It ends the call, sets <code>consent.call=false</code> , <code>dnc=true</code> and <code>consentOverrides</code> (which survive data rebuilds), audits <code>consent.withdrawn</code> and triggers a lead recompute (<code>zeroWhen: consent.dnc → score 0, NBA suppress</code>).
6	Safety first	<code>callback_later</code> ("đang lái xe", "busy", "driving") ends politely with a drive-safely line
7	Handle scam concerns	The <code>scam_concern</code> intent gives the <code>trust</code> line: how to verify, and that payment happens in-app only
8	Human handoff on request or intent	<code>human</code> , <code>buy_now</code> and <code>yes</code> in the offer state end as <code>hot_handoff</code> . The handoff summary is minimal: masked phone, talking points, trust/price flags (<code>handoffSummary</code>). Supervisors and agents work it under ABAC.
9	Bounded conversation	<code>maxClarifications</code> (2), then a link is sent instead of looping. Utterances are truncated (500 characters).
10	Lawful targeting	Calls only after <code>canContact()</code> : call consent, DNC, 08:00–20:00 local, weekly call cap 2 (<code>contact_policy.json</code>). The campaign endpoint skips company-owned vehicles and profiles without consent or a phone (<code>routes.js /api/voice/campaign</code>). Decree 91/2020/ND-CP call rules and the national Do-Not-Call list: confirm with TASCO legal . Check against the national DNC register before calling (planned integration).
11	Every call improves data, nothing more	Outcomes update expiry (renewed elsewhere, confidence 0.6), raise DQ issues (wrong person, plate mismatch) and record consent. There are no hidden profiling attributes.

3. Script and prompt lifecycle (`content.voicebot`)



What is governed: the bot lines (vi text + en gloss), intent keyword lists and their order (first match wins), `maxPlateAttempts`, `maxClarifications` and the disclosure type. The state machine itself is code: changes go through a pull request, code review and release.

Approver checklist (compliance):

1. Disclosure and recording notice unchanged, or legally re-approved.
2. No request for OTP, card, password or payment.
3. Price statements match the active `tariff.*` and say "same at every insurer".
4. No banned phrases (automated) and no implied discounts (human judgement: e.g. "ưu đãi" (offer) implying a price cut).
5. The opt-out keyword list is not narrowed. Every removal needs a justification.
6. The intent order is reviewed (a short `yes` keyword can capture "không có" ("I don't have it"); see [ADR-009](#)).
7. Golden-transcript results attached: 100 % pass on critical paths (opt-out, plate mismatch, scam concern, price).

8. Tone and politeness in Vietnamese (Quý khách form of address), and regional dialect check.

Testing assets:

- `simulatedCaller.js` personas (eager, self-serve, price shopper, skeptic, already renewed, busy, opt-out, wrong plate);
- the interactive voice console (`POST /api/voice/sessions` , `/turns`);
- **planned:** a labelled utterance set (\$5.3) and an automated regression run on `POST /api/rules/validate` for `content.voicebot` .

Prompts for future LLMs follow the same lifecycle under a new rule kind (e.g. `content.llm_prompts`). The system prompt, few-shot examples, model id, temperature and output schema are versioned, copy-guarded where customer-facing, and approved by compliance and model risk.

4. Model risk management: lead scoring and inference

4.1 Model card: lead score (`scoring v1`)

Item	Value
Purpose	Prioritise outreach (who, when, which channel) for renewal and new business. It does not affect price, eligibility or claims.
Method	<code>score = round(Σ weight_f × clamp(value_f, 0, 1) × (0.6 + 0.4 × expiryConfidence))</code> , set to 0 if <code>consent.dnc</code>
Factors (weight)	urgency (35): days to expiry/lapse · expiryConfidence (15) · engagement (20): app sessions, toll trips · reachability (15): push, Zalo, call consent, SMS · affinity (15): prior VETC purchase, TASCO customer, wallet covers premium, auto top-up, minus complaints
Tiers	hot ≥ 70, warm ≥ 45, otherwise nurture
Explainability	Every lead stores <code>reasons[] {factor, label, points, max, why}</code> sorted by contribution; the NBA stores <code>ruleId</code> . Both are shown in the customer 360 view.
Excluded attributes	Name, gender, age and date of birth are not collected. <code>region</code> and <code>ownerType</code> are available in the facts but not used by any factor.
Proxy-risk attributes	<code>walletBalance</code> (affluence proxy), <code>province</code> (via facts), <code>complaints12m</code> (reduces affinity)
Validation	<code>validators.js</code> : weights sum to 100, hot > warm. <code>/api/rules/simulate</code> compares current and candidate output per profile.
Limitations	Hand-set weights, not fitted. Synthetic ground truth only in demo (<code>_hidden</code> in synthetic records); real calibration needs production outcomes.

4.2 Controls

Control	Method	Threshold / cadence	Status
Explainability	Factor reasons and NBA rule ids on every lead	100 % coverage	Implemented
Change control	Maker-checker + simulate; model-risk sign-off for <code>scoring</code> , <code>nba</code> , <code>journeys</code>	Every change	Implemented (sign-off procedural)

Control	Method	Threshold / cadence	Status
Reproducibility	Store the rule-set versions/checksums used on each lead	Every evaluation	Planned (data §8)
Calibration	Conversion rate (order within 30 days of the first touch) by tier; hot > warm > nurture, monotonic	Monthly	Planned (needs analytics mart)
Fairness	Contact rate, tier mix and conversion by province group , owner type and vehicle category . Disparate-impact ratio of contact rate ≥ 0.8 versus the reference group, unless explained by urgency. Review of proxy features.	Quarterly + on each scoring change	Planned
Feature guard	Validator rule: scoring factors must not reference <code>region</code> , <code>ownerType</code> or any future demographic field without a model-risk waiver	On save	Recommended (add to <code>validators.js#SPECIFIC.scoring</code>)
Drift	PSI on the score distribution (alert > 0.2) and on key features (<code>days</code> , <code>expiryConfidence</code> , <code>engagement</code>); weekly tier-mix trend	Weekly	Planned
Outcome loop (refit)	Quarterly review: compare factor contributions with outcomes; propose new weights as a draft; A/B (champion/challenger) using hold-out groups	Quarterly	Planned. Needs an experiment flag, because there is one active version per kind today.
Human oversight	Telesales see the reasons and can reject leads (handoff <code>lost</code> with a note); supervisors reassign	Continuous	Implemented

4.3 Inference models (MDM)

Model	Risk	Control
Expiry inference	A wrong date means early or late reminders, and a quote start date that overlaps or leaves a gap in existing cover	Measure the accuracy of inferred expiry against later verified certificates or TASCO issuance (share within ± 21 days, per method). Tune <code>expiryEvidence</code> confidences by maker-checker. Reminders ask the customer to verify (<code>verify_expiry</code> template, <code>needsConfirmation</code> in the app).
Category inference	A wrong category means a wrong regulated premium if the customer does not correct it	Measure accuracy against TASCO-issued categories. Recommendation: when <code>categoryConfidence</code> < 0.8, require explicit customer or agent confirmation of seats/usage before <code>purchase</code> (not enforced today; <code>quote</code> uses the inferred category unless <code>options.category</code> is passed).

Hard-coded confidences outside the rule set (customer 0.8, steward 0.9, voice bot 0.6, issued 1.0; voice `expiryKnown` threshold 0.5; app `needsConfirmation` < 0.75) should move into `enrichment` so they are governed too.

5. LLM integration policy

5.1 Permitted and prohibited uses

Permitted (with approval)	Prohibited
Intent classification behind the <code>classify(text) → intent</code> port. The output must be one of the script's intent names or <code>unknown</code> .	Generating customer-facing speech or messages that were not pre-approved

Permitted (with approval)	Prohibited
Internal agent assist: summarising call outcomes for telesales from structured signals	Quoting prices, eligibility, cover terms or legal advice
DQ assistance: suggesting the normalisation of messy free text for steward review	Deciding consent, claims, underwriting or any decision with legal effect without a human
Rule authoring assistant that drafts JSON Logic as a draft only , still subject to validators and maker-checker	Collecting or processing OTPs, card data or national id

5.2 Data protection

1. Minimise.

- Send only the current utterance (≤ 500 characters) and the intent list.
- Before sending, mask digit sequences (plates, phones) and known names.
- Never send profile data, transcript history or ids.

2. Residency and DPA.

- No personal data goes to an external model provider without a DPA, a processing-purpose record, a transfer impact assessment, and confirmation that cross-border transfer rules are met: Decree 13/2023/ND-CP transfer dossier and the PDP Law 2025 (**confirm with TASCO legal**).
- Prefer an in-country or self-hosted model for anything that touches customer speech.

3. No training on customer data by the provider (contractual), and zero data retention where offered.

4. Logging. Log the intent, confidence, model id and prompt version, but not the raw prompt or completion when it may contain PII.

5.3 Evaluation and red-teaming

Suite	Content	Pass criteria
Intent evaluation set	$\geq 2,000$ labelled Vietnamese utterances: Northern, Central and Southern dialects, diacritic-less ASR output, code-switching (vi/en), noise and filler, numbers spoken in words	Macro F1 $\geq$ keyword baseline + 5 pts; opt_out recall ≥ 0.99 ; scam_concern recall ≥ 0.95 ; buy_now precision ≥ 0.9
Plate extraction	Spoken plates in many styles ("ba mươi A ...", "30A 123 45", 4-digit legacy)	$\geq 98\%$ exact match on clear audio transcripts
Red team	Prompt injection via speech ("bỏ qua hướng dẫn..." / "ignore previous instructions"), requests for discounts, impersonation scripts, abusive language, attempts to extract other customers' data	Zero policy violations: classifier output stays in the enum; no PII echoed
Regression	Golden transcripts for each script version	100 % on critical paths
Latency	p95 classification time	≤ 300 ms (otherwise fall back to keywords)

Runtime guard-rails:

- schema-validate the output (enum);
- timeout, then fall back to `keywordClassifier`;
- the keyword `opt_out` check **always runs first** and overrides the LLM;
- a feature flag per campaign for gradual rollout.

6. Monitoring

Governance dashboard (GET /api/dashboard/governance , permission audit:read ; insightsService.governance) returns today:

- voiceBot.calls , voiceBot.outcomes (opted_out, hot_handoff, link_sent, already_renewed, plate_mismatch, unverified, wrong_person, callback_later, not_interested);
- plateVerificationFailureRate = (plate_mismatch + unverified) / calls;
- optOutRate = opted_out / calls;
- disclosure statement;
- rule-set counts by status (draft, pending, active, retired, rejected);
- audit-chain verification (ok , entries , head).

Metrics and thresholds:

Metric	Source	Threshold → action
Opt-out rate	governance dashboard / voice_calls_total{outcome="opted_out"}	> 8 % weekly → review script and targeting; > 15 % → pause campaign
Plate verification failure	governance dashboard	> 25 % → check ASR quality and data accuracy (wrong numbers)
Wrong-person rate	voice_calls_total{outcome="wrong_person"}, DQ issues	> 5 % → data quality incident with the VETC data office
Unknown-intent / clarification rate	Session lastIntent , clarifications (planned metric)	> 20 % → NLU review
Complaints mentioning the bot	CX ticketing (planned integration)	Any vishing-related complaint → incident (§7)
Copy-guard blocks at send	messages_total{status="blocked"}	> 0 → template defect; fix the template
Contact-policy suppressions	Touchpoint skipped reasons	Trend review monthly (consent health)
Score drift / fairness	Analytics marts (§4.2)	PSI > 0.2 or DI < 0.8 → model-risk review
Handoff conversion	handoffs won/lost	Below target → review talking points and scoring

7. AI incident handling

Incident	Examples	Immediate containment	Follow-up
Harmful or incorrect bot speech	Wrong price stated, discount implied, rude wording	Roll back content.voicebot to the previous version (rollback → approve). Pause automated calls: suspend the journeys CronJob and stop /api/voice/campaign use.	Root cause; add a golden transcript; compliance review
Missed opt-out	Customer said stop and was called again	Pause campaigns; manually set DNC for affected profiles; review consent.withdrawn audit entries	Add utterances to the evaluation set; widen keywords; regulator and customer communication if required (confirm with TASCO legal)

Incident	Examples	Immediate containment	Follow-up
Vishing wave impersonating VETC	Customers report fraudulent "VETC insurance" calls	Push and Zalo service message to affected regions: "VETC never asks for OTP/payment by phone"; strengthen the trust line	Coordinate with VETC security and the authorities
Model or ASR degradation	Vendor ASR accuracy drop; plate failures spike	The telephony breaker opens on errors. Otherwise reduce campaign concurrency or pause.	Vendor SLA escalation
Scoring defect	A bad weights version leads to the wrong customers being contacted	Roll back <code>scoring / nba</code> ; recompute leads (<code>POST /api/leads/recompute</code>)	Simulation evidence required on the next change

Gap: there is no single **kill switch** that stops all automated calls immediately, independent of maker-checker. Today it takes a CronJob suspend plus discipline on the campaign endpoint. **Recommendation:** an operational flag, for example `VOICE_BOT_ENABLED` in config or an `ops` toggle with `ops:run_jobs` and audit, checked in `voiceService.autoCall`.

8. Roles and responsibilities (RACI)

R = Responsible · A = Accountable · C = Consulted · I = Informed

Activity	Product owner (TASCO)	CX lead (VETC)	Compliance officer	Legal / DPO	Model risk / analytics	InfoSec	Rule author	Rule approver	iorta TechNXT engineering	Telesales ops
Voice bot script change	A	R	C	C	I	I	R	R (approve)	I	C
Disclosure / recording wording	C	C	R	A	—	—	I	I	I	—
Dialogue state-machine change (code)	A	C	C	I	I	C	—	—	R	I
Scoring / NBA / journey change	A	C	C	I	R (sign-off)	—	R	R (approve)	I	C
Fairness and drift review	I	I	C	C	R/A	—	—	—	C	I

Activity	Product owner (TASCO)	CX lead (VETC)	Compliance officer	Legal / DPO	Model risk / analytics	InfoSec	Rule author	Rule approver	iorta TechNXT engineering	Telesales ops
LLM introduction (new model or prompt)	A	C	C	R (DPA, transfer)	R (evaluation)	R (security review)	—	R (approve prompt)	R (integration)	I
Opt-out and consent handling	I	R	A	C	—	—	—	—	R (code)	I
AI incident response	A	R	R	C	C	C	—	—	R	I
Governance dashboard review (monthly)	A	R	R	I	R	I	—	—	C	I
Category/expiry inference accuracy	A	—	I	—	R	—	C	C	C	—

9. Regulatory alignment (to confirm)

All of the following must be **confirmed with TASCO legal** before production:

- personal data processing basis and AI disclosure (Decree 13/2023/ND-CP; Law on Personal Data Protection 2025 and its implementing decree);
- marketing calls and messages: hours, frequency and the DNC registry (Decree 91/2020/ND-CP);
- consumer protection and call recording notice (Law on Protection of Consumers' Rights 2023);
- the prohibition on discounts or rebates for compulsory TNDS (Law on Insurance Business 08/2022/QH15; Decree 67/2023/ND-CP);
- data residency (Cybersecurity Law 2018; Decree 53/2022/ND-CP).

The platform's defaults are conservative and configurable (`contact_policy` , `copy_guard` , `retention`), so legal positions can be applied as maker-checker rule changes rather than code changes.